FORCE – MPSC / Competitive Exams Notes

Definition:

Force is a push or pull that changes or tends to change the state of rest or motion of a body.

$$F = m \times a$$

Where: F = Force applied on the body
m = Mass of the body
a = Acceleration produced in the body.

SI Unit: Newton (N)

Definition of 1 Newton: One Newton is that amount of force which, when applied to a body of mass 1 kg, produces an acceleration of $1\ m/s^2$ in it.

Key Exam Points:

- Force is a Vector quantity (has both magnitude and direction).
- The CGS unit of force is Dyne ($1\ N = 10^5$ dynes).
- Force can change the shape or size of a body.

Example: If $m = 5\ kg$ and $a = 2\ m/s^2$, then Force
$F = 5\ kg \times 2\ m/s^2 = 10\ N$.

POWER – Rate of Doing Work

Definition:

Power is the rate of doing work.

$$\boxed{P = \frac{W}{t}}$$

$$\boxed{P = F \times v}$$

Where: P = Power, W = Work done, t = Time taken, F = Force, v = Velocity.

SI Unit: Watt (W)

Definition: 1 Watt = 1 Joule per second

$$\boxed{1 \text{ horsepower (hp)} = 746 \text{ Watt}}$$

Example: A body of mass 50 kg is lifted to a height of 4 m in 10 s. Calculate Power. ($g = 10\ m/s^2$)

$$\text{Work (W)} = mgh = 50\ kg \times 10\ m/s^2 \times 4\ m = 2000\ J$$

$$\text{Power (P)} = \frac{W}{t} = \frac{2000\ J}{10\ s} = 200\ W$$

Key Exam Points:

- Power is a Scalar quantity.
- The commercial unit of energy is kilowatt-hour (kWh).
- $1\ kWh = 3{\cdot}6 \times 10^6\ J$.
- Horsepower (hp) is a non-SI unit used for motors and engines.

FORCE vs POWER – Difference for MCQ

FORCE	POWER
Push or Pull	Rate of Doing Work
Depends on mass & acceleration	Depends on work & time
Formula: $F = ma$	Formula: $P = \frac{W}{t}$
Unit: Newton	Unit: Watt
Changes motion	Measures speed of work done

Trap-based MCQ Statements:

- 1. The rate of doing work is called ____.
- 2. Unit is Newton – Identify the quantity.
- 3. Which quantity depends on time?
  (a) Force (b) Power (c) Both (d) None.

Mixed Concept Example (Lift Question):

Lift moving up with constant velocity 'v' against gravity & friction.

Force required (F) = Total opposing force.

Power (P) = Force × Velocity

Formula: $P = F \times v$

Summary: Power required by lift motor depends on total force and velocity.